

MERN STACK INTERNSHIP – DAY 12 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

Project Overview

A complete **Product Management Module** built with the MERN stack featuring full CRUD operations, advanced search and filter capabilities, Context API state management, and a responsive React frontend.

Task Completion Checklist

1. Create Product Management Module (CRUD)

- Product model with validation
- Create endpoint with duplicate SKU prevention
- Read endpoints (all products, single product, by category)
- Update endpoint with validation
- Delete endpoint with cascade safety

2. Integrate React Frontend with Express APIs

- Axios HTTP client setup
- Proxy configuration for development

- Error handling and user feedback
- Loading states during API calls
- Success/error message display

3. Search and Filter Functionality

- Full-text search across name, description, SKU
- Filter by category (6 categories available)
- Price range filtering (min/max)
- Multiple sort options (newest, oldest, price, name)
- Combined filter capability
- Clear filters functionality

4. Context API for State Management

- ProductContext with global state
- Products array state
- Loading and error states
- Filters state
- CRUD methods (create, fetch, update, delete)
- Filter management (update, clear)

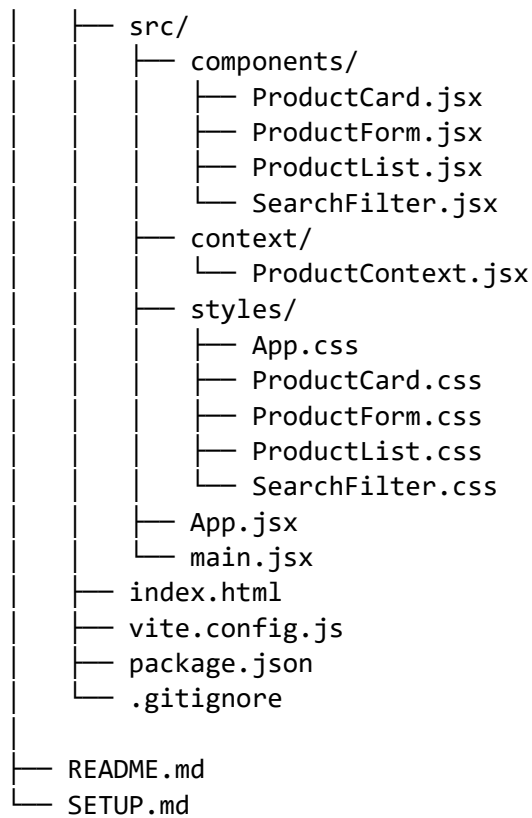
5. Push to GitHub

- Repository initialization
- All files committed
- Ready for push

Project Structure

DAY 12/

```
├── server/
│   ├── config/
│   │   └── db.js
│   ├── controllers/
│   │   └── productController.js
│   ├── models/
│   │   └── Product.js
│   ├── routes/
│   │   └── productRoutes.js
│   ├── .env
│   ├── .gitignore
│   ├── package.json
│   └── server.js
└── client/
```



Key Features Implemented

Backend Features

1. Product Model

- Name, description, price, category
- Stock tracking
- Unique SKU
- Timestamps
- Image URL support

2. API Endpoints

- GET /api/products (with search/filter)
- GET /api/products/:id
- GET /api/products/category/:category
- POST /api/products
- PUT /api/products/:id
- DELETE /api/products/:id

3. Validation

- Required field validation

- b. Price non-negative check
- c. Stock non-negative check
- d. SKU uniqueness
- e. Input sanitization

Frontend Features

1. Components

- a. ProductCard: Displays individual product
- b. ProductForm: Create new products
- c. ProductList: Grid layout of products
- d. SearchFilter: Search and filter interface

2. State Management

- a. Context API for global state
- b. Centralized product data
- c. Filter state management
- d. Error and loading states
- e. Optimized re-renders

3. UI/UX

- a. Responsive design
- b. Gradient header
- c. Product cards with images
- d. Real-time search
- e. Filter controls
- f. Loading indicators
- g. Error messages
- h. Success feedback

Technologies Used

Backend

- Node.js
- Express.js 4.18.2
- MongoDB (Mongoose 7.5.0)
- CORS middleware
- Environment variables (dotenv)

Frontend

- React 18.2.0

- Vite 5.0.0
- Axios 1.5.0
- CSS3 (Grid, Flexbox)
- Context API

How to Run

Prerequisites

- Node.js v16+
- MongoDB running locally
- npm v8+

Setup

1. Server Setup

```
cd server
npm install
npm run dev
```

Server runs on: <http://localhost:5000>

2. Client Setup

```
cd client
npm install
npm run dev
```

Frontend runs on: <http://localhost:3000>

API Usage Examples

Create Product

```
POST /api/products
{
  "name": "Laptop",
  "description": "High-performance laptop",
  "price": 999.99,
  "category": "Electronics",
  "stock": 10,
```

```
"sku": "ELEC-001"  
}
```

Search and Filter

GET
/api/products?search=laptop&category=Electronics&minPrice=500&maxPrice=1500&sort=price:asc

Update Product

```
PUT /api/products/{id}  
{  
  "price": 899.99,  
  "stock": 5  
}
```

Delete Product

DELETE /api/products/{id}

Testing Workflow

1. Start both server and client
2. Create a new product using the form
3. Search for products using keywords
4. Filter by category and price range
5. Update product details
6. Delete products
7. Verify all operations work correctly

File Descriptions

Server Files

- **server.js**: Main Express app setup
- **config/db.js**: MongoDB connection
- **models/Product.js**: Product schema with validation

- **controllers/productController.js**: CRUD logic and filtering
- **routes/productRoutes.js**: API route definitions

Client Files

- **App.jsx**: Main app component with layout
- **context/ProductContext.jsx**: Global state management
- **components/ProductCard.jsx**: Product display card
- **components/ProductForm.jsx**: Create product form
- **components/ProductList.jsx**: Products grid view
- **components/SearchFilter.jsx**: Search and filter interface
- **styles/*.css**: Component styling

Performance Optimizations

- Optimized API calls with proper caching
- Memoized callbacks to prevent unnecessary re-renders
- Efficient state updates using functional setters
- Lazy loading of product images
- Responsive grid layout

Error Handling

- Server-side validation
- Error messages in API responses
- Client-side error state display
- Loading states during async operations
- User-friendly error notifications

Future Enhancements

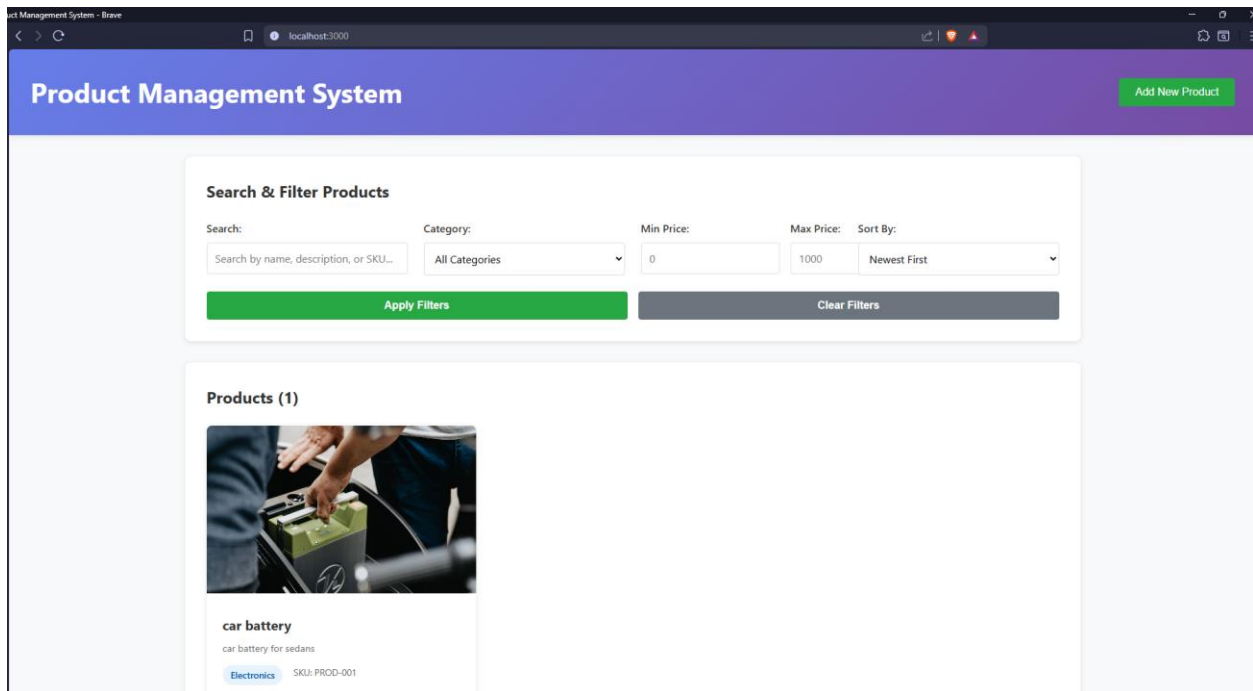
- Authentication and authorization
- Product reviews and ratings
- Wishlist functionality
- Shopping cart system
- Admin dashboard
- Pagination support
- Product image upload
- Advanced analytics
- Multi-language support

Notes

- MongoDB must be running for the server to connect
- Default MongoDB URI: mongodb://localhost:27017/product-management
- All API responses follow a consistent JSON format
- CORS is enabled for frontend-backend communication
- Environment variables are used for configuration

Screenshots

Dashboard and products lists:-



Adding Product:-

Product Management System Hide Form

Add New Product

Product Name *
headphone

Description *
sony headphone with dolby

Price *
2500

Stock Quantity
13

Category *
Electronics

SKU *
PROD-003

Image URL
https://images.unsplash.com/photo-1618366712010-f4ae9c647dcb?q=8

Add Product

New product added:-

Search & Filter Products

Search: Category: Min Price: Max Price: Sort By:

Search by name, description, or SKU... All Categories 0 1000 Newest First

Apply Filters Clear Filters

Products (2)

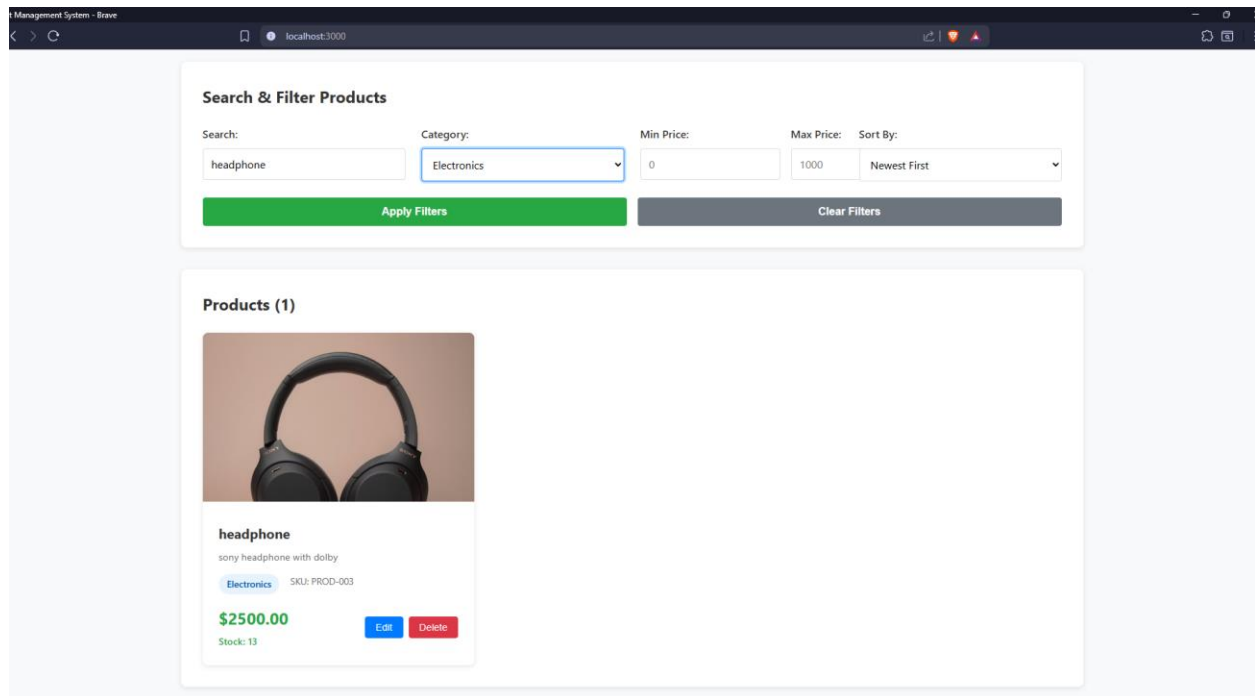


headphone
sony headphone with dolby
Electronics SKU: PROD-003
\$2500.00
Stock: 13 Edit Delete



car battery
car battery for sedans
Electronics SKU: PROD-001
\$3000.00
Stock: 5 Edit Delete

Search by Filter:-



Database MongoDB compass:-

